Seasonal variation of nitrate in shallow wells of the Oued Sabra alluvial aquifer

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Abstract

Nitrate concentrations were measured monthly in 31 hand-dug and drilled wells over two hydrological years. Shallow wells showed a sharp rise in nitrate after the first heavy rains, while wells deeper than 20 m remained stable. The results point to rapid leaching from latrines and livestock enclosures close to the wellheads.

Keywords: groundwater quality; nitrate; alluvial aquifer; seasonal leaching; rural water supply

1. Introduction

Most villages along the Oued Sabra depend on shallow wells for drinking water throughout the year. Nitrate above the guideline value of 50 mg/L is a known risk for infants and is often linked to sanitation close to the well. Earlier surveys sampled only once a year, at the end of the dry season, and may therefore have missed the period of greatest risk.

This note reports two years of monthly monitoring and examines how nitrate changes with season and well depth.

2. Study area and methods

2.1 Sampling

The aquifer consists of sands and gravels up to 45 m thick, recharged mainly by infiltration from the wadi bed during floods. Thirty-one wells were selected to cover three depth classes and a range of distances from the nearest latrine. Samples were collected between 07:00 and 10:00, filtered on site and kept at 4 °C until analysis.

2.2 Analysis

Nitrate was determined by ion chromatography within 48 hours of collection. Field blanks and duplicate samples were included in every batch[a]. Rainfall was recorded daily at the institute's station in Tin Adar.

3. Results

Mean nitrate concentrations are summarised in Table 1. In wells shallower than 10 m, the wet-season mean was 59% higher than the dry-season mean.

Depth class	Dry season	Wet season	Wells
< 10 m	38.4	61.2	14
10–20 m	27.9	41.5	11
> 20 m	12.6	15.3	6

Table 1. Mean nitrate concentration (mg/L) by season and depth class.

Peak values occurred two to four weeks after the first rain event exceeding 25 mm. Six of the fourteen shallow wells exceeded 50 mg/L at least once during the wet season, compared with none of the deep wells.

Distance to the nearest latrine explained more of the variation between wells than distance to livestock enclosures.

4. Discussion

The rapid response of the shallow wells suggests preferential flow through the coarse upper sands rather than slow diffuse recharge. Deep wells are protected by a silty layer at 15 to 18 m, which delays and dilutes infiltrating water. Annual sampling at the end of the dry season would have underestimated the exposure of households using shallow wells.

5. Conclusions

(1) Nitrate in shallow wells varies strongly with season and peaks soon after the first heavy rains.

(2) Monitoring programmes should include at least one sampling round in the early wet season.

(3) Latrines should be sited at least 30 m from wells and, where possible, downslope of them.

References

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[a] Blank values were below the detection limit of 0.2 mg/L in all batches.